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Studierichting: Informatica

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$$\begin{aligned}\int \frac{1}{2 + \cos x} dx &\stackrel{t=\tan\left(\frac{x}{2}\right)}{=} \int \frac{1}{2 + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt = 2 \int \frac{1}{2(1+t^2) + 1 - t^2} dt \\&= 2 \int \frac{1}{t^2 + 3} dt \\&= \frac{2}{\sqrt{3}} \operatorname{Bgtan} \left(\frac{t}{\sqrt{3}} \right) + c \stackrel{t=\tan\left(\frac{x}{2}\right)}{=} \frac{2}{\sqrt{3}} \operatorname{Bgtan} \left(\frac{\tan\left(\frac{x}{2}\right)}{\sqrt{3}} \right) + c\end{aligned}$$

References